

TELANGANA STATE- 2015
CLASS 10
GENERAL SCIENCE, PAPAR – I
(Physical Sciences)

Time: 2 hours 45 min.

Maximum Marks: 40

Instructions:

- (i) The question paper comprises of Four sections - I, II, III and IV.
- (ii) All the questions are compulsory.
- (iii) There is no over-all choice. However there is internal choice to the questions under section-I.
- (iv) In the time duration of 2 hrs. 45 minutes, 15 minutes of time is exclusively allotted to read and understand the question paper.

SECTION -I

(4x4=16 marks)

Note :

- (i) Answer all the questions.
- (ii) Each question carries FOUR marks.
- (iii) There is internal choice for each question. Only one option from each question is to be attempted.
- (iv) Answer each question in 8 to 10 sentences.

1. An object is placed at the following distances from a convex lens of focallength 10 cm.

- (a) 8 cm
- (b) 15 cm
- (c) 20 cm
- (d) 25 cm

Which position of the object will produce?

- (i) A diminished, real and inverted image? 25cm
- (ii) A magnified, real and inverted image? 15cm
- (iii) A magnified, virtual and erect image? 8cm
- (iv) An image of same size as the object? 20cm

Justify your answer in each case.

Ans.

- (a) Case1: Object at 8cm from the lens

In this case, a magnified, virtual and erect image is produced because the object is placed at a distance less than the focal length (10cm).

- (b) Case2: Object at 15cm from the lens

In this case, a magnified, real and inverted image is produced because the object is placed between F and $2F$.

- (c) Case3: Object at 20cm from the lens

In this case, an image of same size as the object is produced because the object is placed at $2F$ and Image is formed at $2F'$.

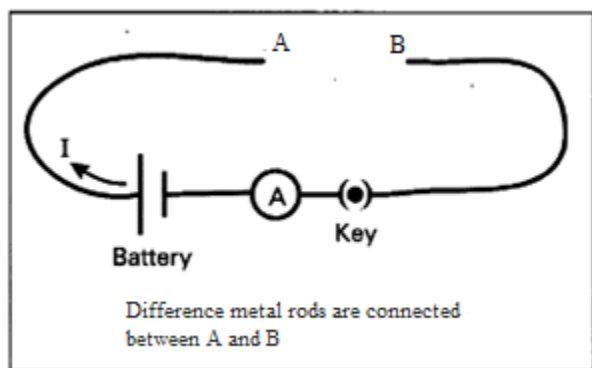
(d) Case 4: Object at 25cm from the lens

In this case, a diminished, real and inverted image is produced because the object is placed beyond $2F$.

OR

1. How do you verify that resistance of a conductor of uniform crosssection area is proportional to the length of the conductor at constant temperature?

Ans.



Procedure:

1. Create a circuit as given in image above – Leave a gap between A and B and insert Ammeter and Plug Key. Connect iron spokes and see the value of current. With increase in length there will be a decrease in current. Thus the resistance of each spoke increases with increase in the length for a constant potential difference.

From the above method, we conclude $R \propto L$

We proved, resistance was more when the cross - section area was less.

Therefore , $R \propto L/A$

Thus we verify l and A

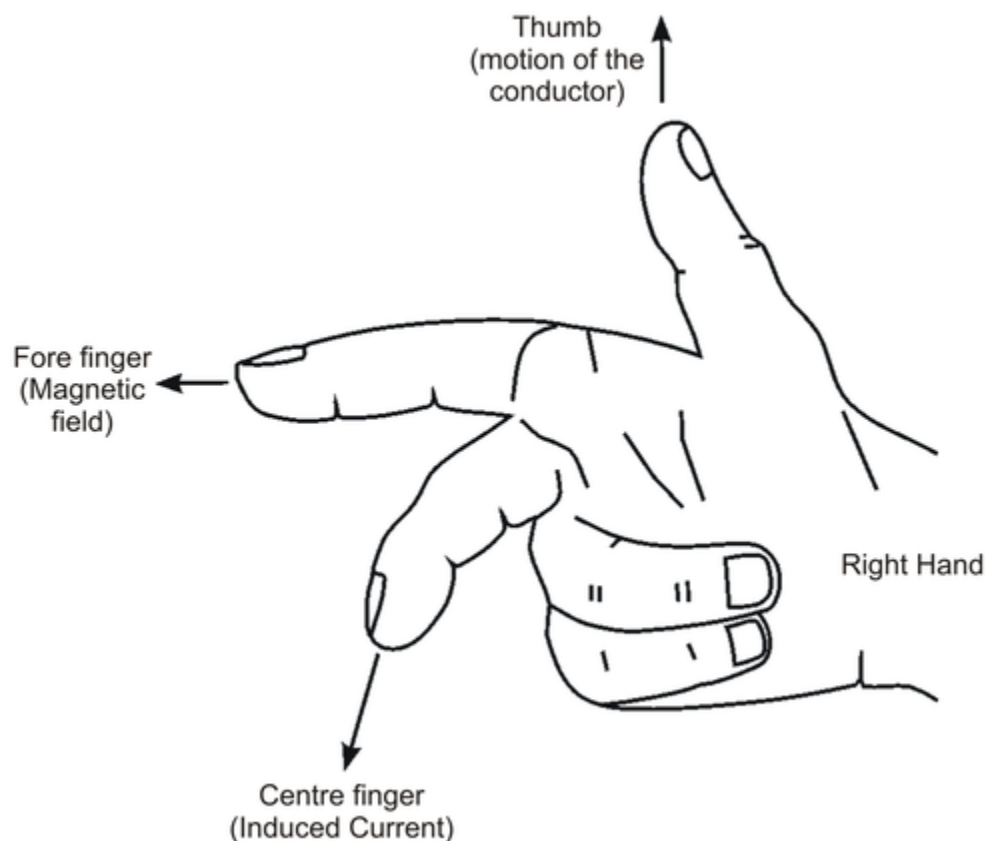
2. Explain the process of melting and latent heat of fusion.

Melting - Melting is the change of state from a solid to a liquid. Melting of a pure substance occurs at a particular constant temperature called melting point. In order to melt a solid substance, you have to add heat to it. When you do this, the molecules in the substance start getting excited and begin to move around more. This moving around makes it difficult for the particles in the substance to hold on to each other as tightly as they do in the solid phase, so they break apart and the substance turns to liquid.

Latent Heat of Fusion - We can define the Latent Heat of Fusion as the amount of heat required to change a unit mass from solid to liquid at a constant temperature. In general, it is the heat needed to melt the substance from solid or the amount of heat to be removed from a unit substance to freeze it from liquid to solid.

OR

2.(a) State Right hand rule with a labelled diagram.



As per the image, Fleming's Right Hand Rule can be explained as -

Thumb: direction of the movement of conductor

Fore-finger: direction of the magnetic field

Middle finger: direction of the induced current.

This rule can be applied on generators and dynamos.

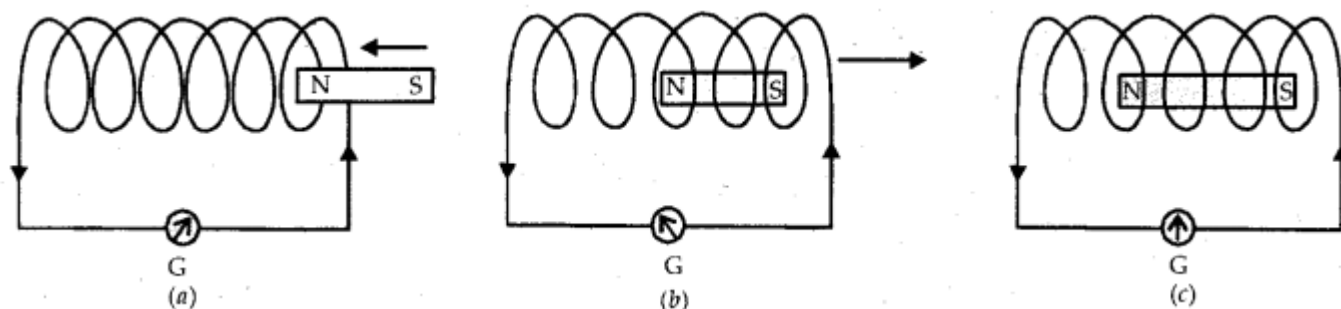
(b) A coil of insulated Copper wire is connected to a Galvanometer.

What happens, if a bar magnet is

(a) Pushed into the coil?

(b) Withdrawn from inside the coil?

(c) Held stationary inside the coil?



Assuming the North side of the bar magnet faces the coil:

(a) Pushed into the coil?

The galvanometer shows deflection. The galvanometer points in the direction of the bar magnet.

(b) Withdrawn from inside the coil?

The galvanometer shows deflection. The galvanometer points opposite to the direction of the bar magnet.

(c) Held stationary inside the coil ?

The Galvanometer shows no deflection.

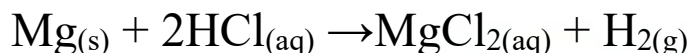
3.(a) Equal lengths of Magnesium ribbons are taken in two test-tubes X and Y. Hydrochloric acid is added to test-tube X and Acetic acid is added to test-tube Y.

**In which test-tube, the reaction will be more vigorous?
Why?**

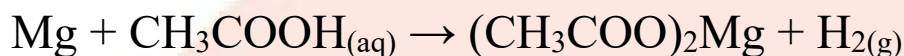
Fizzing is produced when acid reacts with magnesium metal.

Eg - $\text{Mg} + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{H}_2$

In test tube X:



In test tube Y:



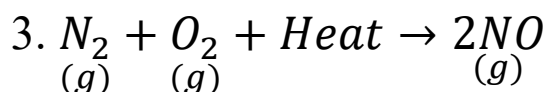
Since HCL being a stronger acid so the reaction will be more vigorous in Test Tube X

(b) Name the four chemicals that are obtained from common salt and write their molecular formulae.

Four chemicals that are obtained from NaCl (common salt) include:

- 1) CaOCl_2 - Calcium oxychloride (bleach)
- 2) NaHCO_3 - Sodium bicarbonate (baking soda)
- 3) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ - Sodium carbonate with 10 atoms of water (washing soda)
- 4) NaOH - Sodium hydroxide

OR



**What information do you get from the above equation?
Comment. Write an activity about how you conduct an**

experiment to show that more reactive metals replace less reactive metals from their compounds.

Ans. The above reaction is an endothermic reaction and an addition reaction.

PROCEDURE:

Two nails were taken and was rubbed with sandpaper so that they look greyish. Two test tubes were taken and copper sulphate was added. A nail was dipped in one of the test tubes containing copper sulphate solution. Another nail was left in a dish as it is.

OBSERVATION:

The test tube containing copper sulphate solution turned into light green due to presence of nail in the test tube. Also the nail dipped in test tube has a brown coating of copper. The nail in the dish remains as it is since it was not exposed to copper sulphate solution. Also the copper sulphate solution color in the second test tube also remains the same since there was no reaction due to absence of nails in the test tube.

INFERENCES:

Green Color – Shows presence of Fe^{2+} ions. So iron is more reactive than copper

Brown Color of nail – Shows presence of copper in the nail instead of iron.

4. Two elements X and Y belong to Groups 1 and 2 respectively in the same period of the Periodic Table. Compare these elements with respect to:

- (i) number of electrons in their outer-most orbit.
- (ii) their atomic size and their valencies.
- (iii) their ionisation energy and metallic character.
- (iv) formulae of their chlorides and sulphates.

Ans. Given Information:

- X and Y belong to same period
- X belongs to group '1'
- Y belongs to group '2'.

(i) Since X belongs to group 1 and Y belongs to group 2 and
Valence electrons in an atom = Group number

So, the Valence electron in element X = 1

valence electrons in Y = group number = 2.

(ii) Valence Electron of X = 1. To get to stable gas configuration the elements require 1 electron. So, the valency of X = 1 whereas valence electrons in element Y = 2 and is already stable. So, valency of Y is 2.

The size of the atom decreases when move from left to right in a period. So, the size of X will be larger than Y and is 4 more than the size of Y.

(iii) X will have more metallic character than Y since Metallic character decreases on moving from left to right in a period and X lies in group 1 and Y in group 2.

(iv) The Sulphates of element X and Y are:

Oxide of X is X_2O

Oxide of Y is YO

Valency of X is 1. So, Chloride of X = $XC\ell$

Valency of Y is 2, Chloride of Y = $Y\ell_2$

OR

4. Four metals A, B, C and D are in turn added to the following solutions one by one. The observations made are tabulated below.

Answer the following based on the above information:

Metal	Iron (II) sulphate	Copper (II) sulphate	Zinc sulphate	Silver nitrate
A	No reaction	Displacement	-	-
B	Displacement	-	No reaction	-
C	No reaction	No reaction	No reaction	Displacement
D	No reaction	No reaction	No reaction	No reaction

(i) Which is the most reactive metal? Why?

(ii) What would be observed, if 'B' is added to a solution of Copper (II) sulphate and why?

(iii) Arrange the metals A, B, C and D in order of increasing reactivity.

(iv) Which one among A, B, C and D metals can be used to make containers that can be used to store any of the above solutions safely?

Ans.

Zn	Most Reactive
B	
Fe	Reactivity Series
A	
Cu	
C	
Ag	
D	Least Reactive

- (i) Since B replaces iron from its salt solution so B is the active metal.
- (ii) B is more reactive so it will replace Cu from CuSO_4 solution
- (iii) $D < C < A < B$
- (iv) Metal D will not replace Zinc and Silver from their salt solution so it should be used to store the ZnSO_4 and AgNO_3 solutions.

SECTION - II

(6x2=12 marks)

Note:

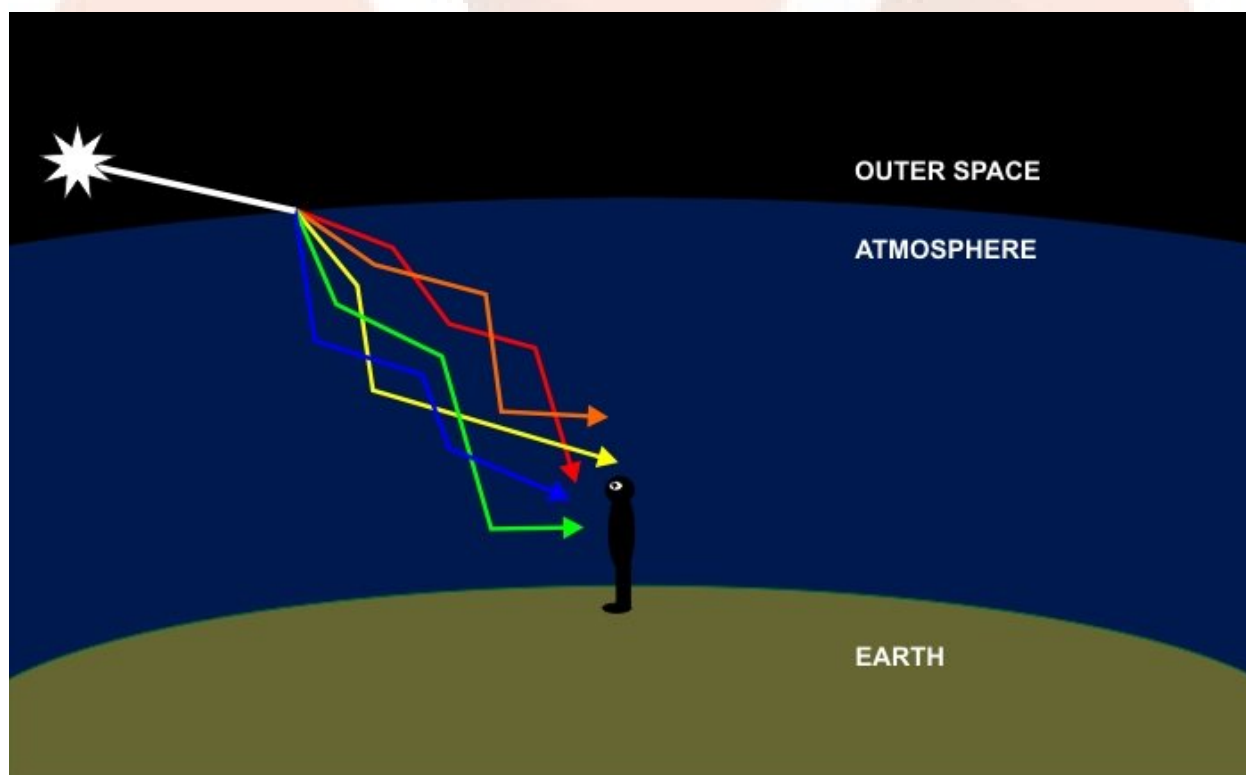
- (i) Answer all the questions.
- (ii) Each question carries TWO marks.
- (iii) Answer each question in 4-5 sentences.

5. Suggest reasons for the phenomenon associated with the following.

(i) The sky appearing blue.

Since this is related to scattering of lights. The intensity of scattered lights is inversely proportional to fourth power of wavelength according to Rayleigh law. There are small particles in sky and when a light is incident then blue color is scattered the most. That's why everyone of us can see the sky as blue.

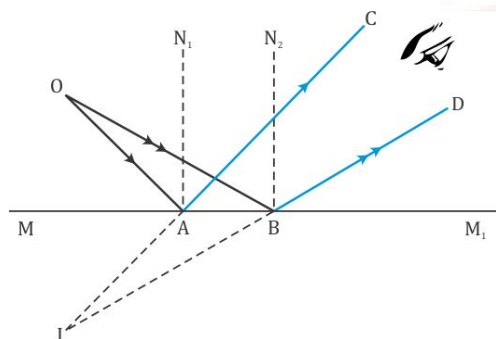
(ii) Twinkling of stars.



Starlight while entering the earth's atmosphere undergoes refraction. Due to this the position of starlight is apparently changed than its original position. Since the actual position of

star is varying it looks twinkling due to change in brightness of light entering the eyes.

6. Draw the diagram that explains the formation of an image by a plane mirror.



7. A ray of light enters from air to a medium X. The speed of light in the medium is 1.5×10^8 m/s and the speed of light in air is 3×10^8 m/s. Find the Refractive index of the medium x.

$$\text{Refractive index of a medium} = \frac{\text{Speed of light in air}}{\text{Speed of light in a medium}}$$

$$\eta = \frac{3 \times 10^8}{1.5 \times 10^8}$$

$$\therefore \eta = 2$$

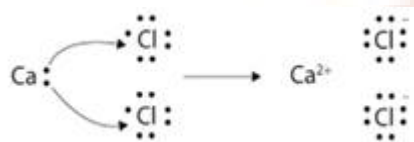
8. For a better understanding about the electronic configuration in an atom, the teacher wrote shorthand notation nl^x on the black-board. Looking at this notation, what could be the probable questions that generate in the student's mind? Write any two of them.

[ans] What is shorthand notation in chemistry?

What is shorthand configuration?

9. Represent each of the following molecules using Lewis notation.

(i) Calcium and Chlorine to form Calcium chloride.



(ii) Formation of Oxygen molecule from Oxygen atoms.



2. (a) Why are vegetable oils healthy as compared to vegetable ghee?

[ans] Since vegetable ghee contains saturated fatty acids which is harmful for the body since it increases cholesterol level in the body where as Vegetable oils contain unsaturated fatty acids which are good for the health

(b) $\text{CH}_3 - \underset{\text{Cl}}{\text{CH}} - \text{CH} = \text{CH}_2$ has the IUPAC name

[ans] 4-Chloropent-1-ene

SECTION – III

(7x1=7 marks)

Note:

(i) Answer all the questions.

(ii) Each question carries ONE mark.

(iii) Answer each question in 1 or 2 sentences.

3. What is the cause of Presbyopia?

It means loss of vision to nearby objects due to aging since the eyes lens becomes harder and it can no longer change the shape and thus loses its power to focus on nearby objects.

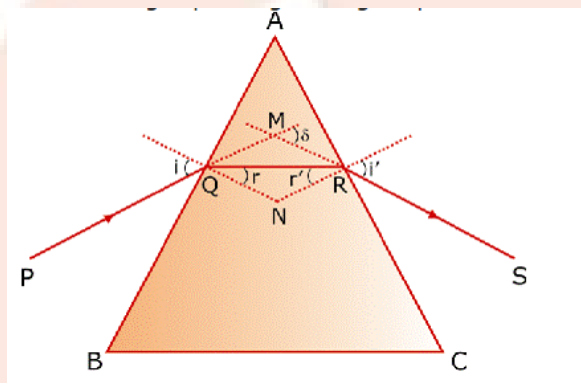
4. Draw a ray diagram to show the angle of deviation

when a ray of light passes through a glass prism.

i - angle of incidence

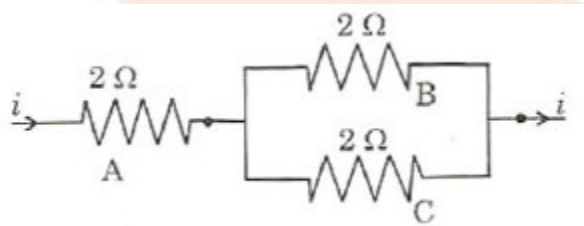
r and r' -angles of refractions

δ - angle of deviation



Consider light passing through a prism ABC as shown

5.



Three resistors A, B and C are connected as shown in the figure. Each of them dissipates energy to a maximum of 18 W. Find the maximum current that can flow through the three resistors.

[ans] V = voltage across a resistor

I = current through the resistor

R = value of resistance.

Power dissipated by the resistor is given by: $P = V^2/R = I^2 R$

$$I^2 = P / R$$

$$I = \sqrt{P/R}$$

$$I = \sqrt{18/2}$$

$$I = 3A$$

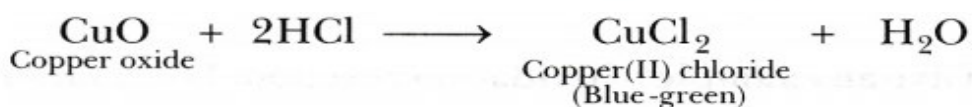
So, max current = 3A

6. What happens when a current carrying wire is placed in a magnetic field?

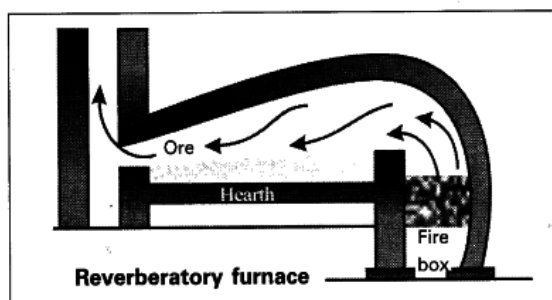
[ans] According to Fleming's left-hand rule. if a current carrying conductor is placed in a magnetic field, it experiences the Lorentz force (unless the angle between the flow of current and magnetic lines is 0°).

7. On adding dilute Hydrochloric acid to Copper oxide powder, the solution formed is blue-green. Write the new compound formed.

[ans] The new compound formed is copper(II) chloride (CuCl_2), which imparts blue-green colour to the solution.



8. Draw a neat labelled diagram of a Reverberatory furnace.



[ans]

9. How do you explain the role of Oxygen in combustion process?

[ans] The main benefit of oxygen is that it's the most commonly available oxidizer. The main component of air that support combustion is oxygen. Oxygen is a great oxidizer because it is so electronegative, which means it really wants to accept electrons.

SECTION - IV

($10 \times \frac{1}{2} = 5$ marks)

Note:

- (i) Answer all the questions.
- (ii) Each question has FOUR choices. Choose the correct answer for each question and write the relevant alphabet (A, B, C, D) against the question number in your booklet.
- (iii) Each question carries $\frac{1}{2}$ mark.

18. The temperature (T) of two samples of the same substance with masses m_1 and m_2 and temperatures T_1 and T_2 , when added together is ...

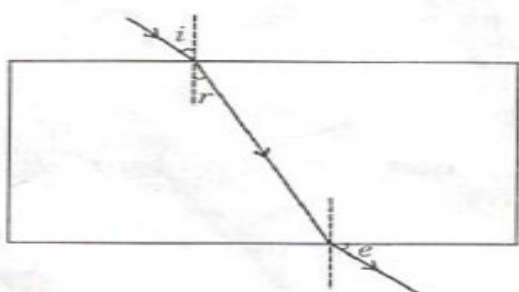
(A) $\frac{m_2 T_1 + m_1 T_2}{m_1 + m_2}$

(B) $\frac{m_1 T_1 + m_2 T_2}{m_1 + m_2}$

(C) $m_1 T_1 + m_2 T_2$

(D) $m_1 T_2 + m_2 T_2$

19. In the diagram, the correctly marked angles are



(A) $\angle i$ and $\angle r$

(C) $\angle r$ and $\angle e$

(B) $\angle i$ and $\angle e$

(D) $\angle i$, $\angle r$ and $\angle e$

[ans] (A) $\angle i$ and $\angle r$

$\angle e$ is the angle of emergent (between normal and angle of emergent)

20. A teacher held a pencil close to a spherical mirror and asked four students W, X, Y and Z to predict the nature of

**the mirror with the help of the image formed in the mirror.
The image was erect and enlarged.**

The four students identified it as follows:

W-Convex in nature

X - Concave in nature

Y- Plane mirror

Z-Plano-concave mirror

The correct statement was given by

(A) W

(B) X

(C) Y

(B) Z

[ans] (C) Y Concave in nature

The Concave mirror will give the erect and enlarge mirror

21. The far point of a myopia eye is 1.5 m. To correct this defect of the eye, the power of lens is...

(A) 0.66 D

(B) -0.66 D

(C) +1.5 D

(D) -1.55 D

[ans] (B) -0.66 D

$$1/f = 1/v - 1/u$$

$$= 1/-150 - 1/-\infty$$

$$= - (1/150) + 0$$

$$= - 1/150$$

$$f = -150 \text{ cm}$$

$$P = 1/f = 1/-m = - 0.66 \text{ D}$$

22. A device for producing electric current is ...

(A) Ammeter

(B) Voltmeter

(C) Generator

(D) Galvanometer

[ans] (c) Generator

electric generator which converts mechanical energy into electricity.

23. When Ethanoic acid is added to a solution of substance X, a colourless and odourless gas Y is liberated. The gas Y turns lime-water milky.

The substance x is...

(A) NaHCO_3

(B) NaOH

(C) CH_3COONa

(D) NaCl

[ans](A) NaHCO_3

Sodium hydrogen carbonate.

When ethanoic acid reacts with sodium hydrogen carbonate or sodium carbonate, colourless and odourless gas carbon dioxide is released which turns lime water milky.

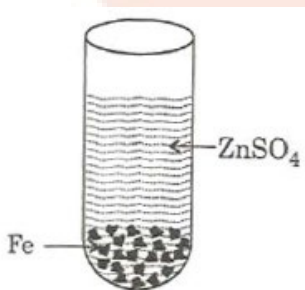
24. Soaps do not create water pollution because

- (A) Soaps are insoluble in water.
- (B) Soaps are soluble in water.
- (C) Soaps are 100% bio-degradable.
- (D) Soaps are non-biodegradable.

[ans] (C) Soaps are 100% bio-degradable.

Soaps are usually biodegradable, so they don't cause water pollution.

25. The correct observation made by the student after putting clean pieces of Iron in the test-tube containing Zinc sulphate are as shown in the figure.



- (A) Solution becomes colourless and Zinc gets deposited on Iron.
- (B) Solution becomes green and Zinc gets deposited on Iron.

(C) Iron pieces get dissolved in the solution making it green.

(D) No reaction is observed.

[ans] (D) No reaction is observed.

Iron is less reactive than zinc which means it is unable to displace zinc from sulphate solution.

26. The maximum number of electrons that can be accommodated in all

the orbitals for which $l = 3$ is ...

(A) 6

(B) 10

(C) 14

(D) 18

[ans] (A) 6

27. Mendeleev's eka-aluminium is

(A) Scandium

(B) Gallium

(C) Germanium

(D) Indium

[ans] (B) Gallium